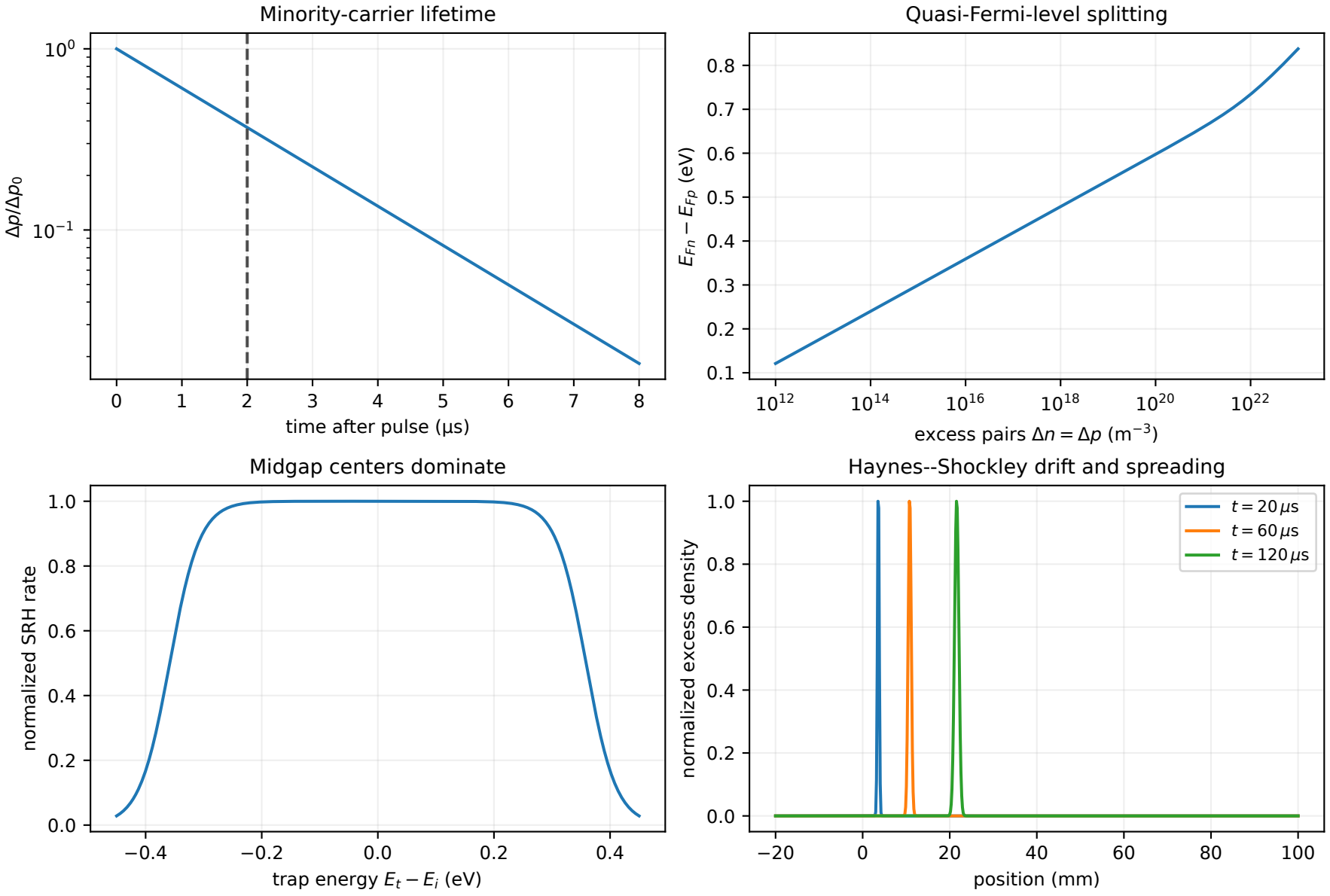
Minority-carrier lifetime
10^0
10^{-1}
$\Delta p/\Delta p_0$
0
1
2
3
4
5
6
7
8
time after pulse (µs)
Quasi-Fermi-level splitting
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
$E_{Fn} - E_{Fp}$ (eV)
10^{12}
10^{14}
10^{16}
10^{18}
10^{20}
10^{22}
excess pairs $\Delta n = \Delta p$ (m^{-3})
Midgap centers dominate
1.0
0.8
0.6
0.4
0.2
0.0
normalized SRH rate
−0.4
−0.2
0.0
0.2
0.4
trap energy $E_t - E_i$ (eV)
Haynes--Shockley drift and spreading
$t = 20\,\mu s$
$t = 60\,\mu s$
$t = 120\,\mu s$
1.0
0.8
0.6
0.4
0.2
0.0
normalized excess density
−20
0
20
40
60
80
100
position (mm)